

PS: Hospital Staff Schedule

Doctors

Algorithm: HOSPITAL STAFF SCHEDULE

INPUT: POP-SIZE, CROSSOVER-RATE, MUTATION-RATE, MAX-GEN,
STAFF-LIST, SHIFTS (Day/Night), Patient-Demand - per shift

OUTPUT: best-schedule

1. Define Fitness (Schedule) :

Penalty $\leftarrow 0$

For each shift in schedule DO

If total-assigned-staff $<$ required-staff-for-patient-demand

Then

Penalty $\leftarrow$ Penalty + 100 // Critical understaffing Penalty

End If

End For

For each staff-member in STAFF-LIST DO

If staff-member works consecutive Day & night shifts then

Penalty $\leftarrow$ Penalty + 50 // Fatigue Penalty

If total-hours-worked $>$ max-allowed-hours. THEN

Penalty $\leftarrow$ Penalty + 50 // overtime Penalty

End For

Return $(1 / (1 + \text{Penalty}))$ // Higher score means fewer Penalties

2. Population $\leftarrow$ random Schedules matching STAFF-LIST & SHIFTS (POP-SIZE)

3. For generation ≥ 1 to MAX-GEN DO

4. Evaluate Fitness of each schedule in Population

5. Update best-schedule if a schedule with a higher fitness is found

6. new-population $\leftarrow$ empty

7. while Size (new-population) $<$ POP-SIZE DO

8. Parent1, Parent2 $\leftarrow$ Select-By-Fitness (Population)

9. If Random() $<$ Crossover-Rate Then

10. Child1, Child2 $\leftarrow$ Crossover - Shifts (Parent1, Parent2)

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11. Else
12. child1, child2 ← Parent1, Parent2
13. End if
14. mutate_shift_assignment(child1), mutate_shift_assignment(child2) with mutation_rate
15. Add child1, child2 to new_population (new_size ← pop_size)
16. End while
17. Population ← new_population
18. End For
19. Return best_schedule
END

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Description: Hospital staff scheduling problem requires designing an optimal shift roster for doctors designing an optimal shift roster for doctors. It must be feasible across days & night shifts. It must balance dynamic patient demand to ensure every shift is safely staffed while strictly complying with legal labor laws & maximum working hours limits.

A genetic algorithm is used to rapidly evolve & discover a high-quality, conflict-free schedule.